

NEWTON'S THIRD LAW – DIRECT CONTACT

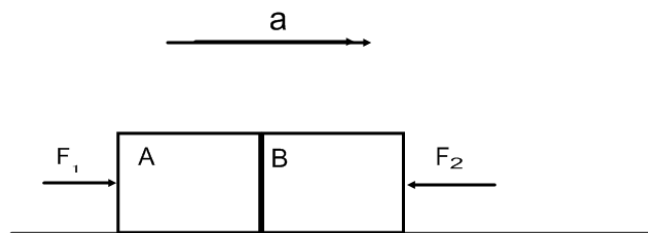
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1. Two blocks, of masses 3 and 9 kg, rest adjacent on a smooth table. I apply a driving force of 36N to the lighter block.
 - a. Draw a force diagrams for:
 - i. the whole system.
 - ii. the lighter block
 - iii. the heavier block
 - b. Calculate the acceleration of the system
 - c. Calculate the magnitude of the force of the lighter block on the heavier block.
 - d. Calculate the magnitude of the force of the heavier block on the lighter block.
2. A block of mass 5 kg rests on a table. A block of mass 2 kg rests on top of the 5 kg block. When an 8 N force is applied to the 5 kg block, the 2 kg block remains in contact with the 5 kg block without slipping. The frictional force between the 5 kg block and the table is 1 N.
 - a. Draw a diagram to show the horizontal forces acting on the whole system (i.e. treat the two blocks as a single object).
 - b. Find the acceleration of the system.
 - c. Draw a diagram to show the horizontal forces acting on the smaller block.
 - d. Hence find the frictional force between the two blocks.

3. A truck of mass 2000 kg is carrying a pallet of mass 250 kg. While the truck is travelling there are resistive forces of 500 N on the truck and 200 N on the pallet. A driving force of 800 N is applied to the truck, and in the resulting acceleration the pallet does not slip.
- Draw a diagram to show the horizontal forces acting on the truck.
 - Draw a diagram to show the horizontal forces acting on the pallet.
 - Write down the equations of motion for both the truck and the pallet.
 - Solve your equations to find the acceleration of the system, and the frictional force between the pallet and the truck.
 - Assume now that the back of the truck is smooth but that the pallet is no longer subject to resistive forces. What will happen to the pallet? Find the new acceleration of this system.
4. A car of mass 1200 kg accelerates from rest to 60 km h^{-1} in 10 s. There is a book of mass 120 g on the back seat of the car. Find the minimum frictional force required between the book and the back seat if the book is not going to slip throughout this motion.
5. A 3 kg block sits on top of a 7 kg block, which is resting on a table. The maximum frictional force between the table and the 7 kg block is 2 N. A 5 N force is applied to the 3 kg block, and in the resulting motion the blocks accelerate together. Find this acceleration, and also the frictional force between the two blocks.
6. In this question model resistive forces as negligible. A van of mass 1600 kg is carrying a bucket of mass 10 kg, and travelling at a constant speed of 24 ms^{-1} . When a small child walks out into the road, 30 m ahead of the van, the van is forced to brake quickly, and it comes to rest just in front of the child. If the maximum frictional force between the bucket and the van is 90 N find:
- the acceleration of the bucket;
 - the braking force on the van;
 - the distance travelled by the bucket during the motion (assuming that it continues to accelerate at the same rate until coming to rest)
7. A cube of mass 6 kg rests on top of a horizontal table. A smaller cube of mass 2 kg is placed on top of the 6 kg cube. Find the force that the larger cube exerts on the smaller cube, and the force that the larger cube exerts on the table.

8. Let F_{AB} be the force of A on B and F_{BA} the force of B on A.

For each of these questions, evaluate all of the unknown quantities by considering A and B as a single particle and separately.



(a) $F_{AB} = ?$ $F_{BA} = ?$ $F_1 = 10 \text{ N}$ $F_2 = 6 \text{ N}$ $M_A = 3 \text{ kg}$ $M_B = 5 \text{ kg}$ $a = ?$

(b) $F_{AB} = 16 \text{ N}$ $F_{BA} = ?$ $F_1 = ?$ $F_2 = ?$ $M_A = 2 \text{ kg}$ $M_B = 4 \text{ kg}$ $a = 3 \text{ ms}^{-2}$

(c) $F_{AB} = ?$ $F_{BA} = ?$ $F_1 = 3 \text{ N}$ $F_2 = 6 \text{ N}$ $M_A = 1 \text{ kg}$ $M_B = ?$ $a = -2 \text{ ms}^{-2}$

(d) $F_{AB} = 2 \text{ N}$ $F_{BA} = ?$ $F_1 = 1 \text{ N}$ $F_2 = 4 \text{ N}$ $M_A = ?$ $M_B = 2 \text{ kg}$ $a = ?$

(e) $F_{AB} = 2 \text{ N}$ $F_{BA} = ?$ $F_1 = 8 \text{ N}$ $F_2 = 1 \text{ N}$ $M_A = ?$ $M_B = ?$ $a = 4 \text{ ms}^{-2}$